

Computational evidence for a distributed AT-based regulatory mechanism enabling de novo gene birth in *Pandoravirus salinus*

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Abstract

Pandoravirus salinus possesses the largest known viral genome (2.47 Mb) with 92.5% of genes classified as ORFans. While de novo gene creation from non-coding sequence has been proposed as the primary diversifying mechanism, the genomic infrastructure enabling this process remains poorly understood. Here we present a 12-module computational framework to characterize de novo gene birth in *P. salinus*.

Cross-genome comparison reveals *P. salinus* as a compositional outlier: the lowest GC coefficient of variation (6.1%) despite being the largest genome, the strongest coding-intergenic GC gap (10.7 pp), and ORFan genes indistinguishable from annotated genes across all codon metrics arguing against patchwork horizontal gene transfer. We identify a five-tier proto-gene continuum with monotonic gradients from established genes (GC 0.650, 1,452 bp) to proto-genes approaching intergenic composition (GC 0.547, 216 bp), supported by 96.8% compositional singletons and ESM-2 structural non-separability ($p = 0.90$).

Poly-A/T signals with regulatory positioning pre-exist throughout intergenic space, producing the strongest gene boundary signals tested (+10.3/+9.8 pp AT enrichment). Strand-aware validation confirms 74% of boundary-associated runs in correct regulatory position ($p < 10^{-6}$). BLASTp analysis reveals 81.8% of annotated and 97.0% of hypothetical genes are Pandoravirus-specific orphans ($OR = 7.19$, $p = 7.79 \times 10^{-12}$).

These findings establish a four-stage maturation model: regulatory signals (pre-existing) -> protein structure -> codon optimization -> GC composition; providing independent support for the de novo gene birth hypothesis.

Keywords: de novo gene birth, ORFan genes, Pandoraviridae, regulatory signals, proto-gene continuum, genome evolution

Significance Statement

Giant viruses with genomes rivaling those of bacteria create most of their genes from scratch rather than acquiring them from other organisms, but how non-coding DNA gains

the ability to function as a gene has remained unclear. We show that the largest known virus, *Pandoravirus salinus*, maintains a system of regulatory signals distributed throughout its non-coding DNA, providing the machinery for gene expression before genes even exist. This resolves a key puzzle, new genes can arise anywhere in the genome because the infrastructure to activate them is already everywhere and reveals a specific evolutionary sequence from non-coding DNA to functional gene: first regulatory signals, then protein structure, then codon optimization, and finally bulk nucleotide composition.

Introduction

The discovery of *Pandoravirus salinus* revealed the largest viral genome known to science 2.47 megabases encoding approximately 1,430 predicted proteins, of which 92.5% have no detectable homologs outside the Pandoraviridae (Philippe et al. 2013). This unprecedented ORFan fraction poses a fundamental evolutionary question: where do these genes come from?

The Claverie group has provided compelling evidence that de novo gene creation from non-coding sequence is the primary diversifying mechanism of the Pandoraviridae. Legendre et al. (2018) conducted strand-specific RNA-seq across four *Pandoravirus* strains, demonstrating that 83-87% of genome sequence is transcribed, that palindromic hairpin structures mark approximately 70% of transcript 3' ends, and that the Pandoravirus pangenome is open each new strain adds roughly 60 previously unseen proteins. Jeudy et al. (2019) provided direct evidence by comparing the nearly identical genomes of *P. celtis* and *P. quercus* (96.7% identity), identifying genes in one strain that correspond to intergenic regions in the other, genes captured in the act of being born. Aherfi et al. (2018) confirmed the open pangenome with a small conserved core, and Bisio et al. (2023) demonstrated through CRISPR/Cas9 knockout that essential genes concentrate at the 5' end while ORFans accumulate at the 3' end of *P. neocaledonia*.

These studies establish what is happening (genes are emerging from non-coding sequence) but leave a mechanistic gap: how does intergenic space acquire the regulatory and coding capacity to produce functional genes? If new genes require promoters and terminators to be expressed, how do they arise in intergenic regions distant from existing regulatory elements? And if gene birth is a gradual process, what is the sequence of events from non-coding DNA to functional protein?

Additional compositional analyses have revealed that *Pandoravirus* genomes are subject to unusual sequence constraints. Poirot et al. (2019) documented the near-complete absence of the AGCT tetranucleotide in some strains, pointing to a novel DNA editing or selection process. Sun et al. (2015) identified miniature inverted repeat transposable elements (MITEs) colonizing the intergenic space of *P. salinus*, demonstrating that multiple forces shape the non-coding landscape from which new genes emerge.

Here we present a multi-method computational framework that addresses the mechanistic gap. Through a 12-module analytical pipeline encompassing genome architecture, null model calibration, codon usage analysis, gene prediction validation, evolutionary classification, gene family detection, protein structure prediction, regulatory signal discovery, transcriptomic validation, and genome-wide homology analysis, we provide four novel contributions. First, we establish that *P. salinus* is compositionally the most uniform genome in a four virus comparison set despite being the largest, with ORFan genes indistinguishable from annotated genes by every metric applied arguing powerfully against recent patchwork horizontal gene transfer. Second, we identify a quantitative five-tier proto-gene continuum with monotonic gradients across all measured features, supported by gene family analysis (96.8% singletons) and protein structure prediction (structural non-separability of proto-genes), providing a maturation metric for gene birth. Third, we discover a distributed AT-based regulatory system that preloads intergenic regions with promoter and terminator signals, resolving the mechanistic gap of how genes arise far from existing regulatory elements. Fourth, we establish an asymmetric maturation ordering regulatory signals pre-exist, followed by structural features, codon optimization, and finally GC composition that defines the temporal sequence of de novo gene birth.

Our approach differs fundamentally from previous work: all findings derive from single-genome computational analysis of *P. salinus* without requiring multi-strain comparison, providing an independent line of evidence that complements the comparative genomics approach of the Claverie group.

Results

Genome-wide compositional landscape

To establish the baseline against which gene birth signatures are evaluated, we characterized the compositional architecture of *P. salinus* using a four-genome comparison ladder spanning three orders of magnitude in genome size (Table 1). *P. salinus* emerged as a compositional outlier: it has the lowest coding density (67.5% vs 88-100% for other genomes), the largest intergenic fraction (32.5%), and by far the strongest GC gap between coding (64.6%) and intergenic (53.9%) regions at 10.7 percentage points compared to 7.0 pp for Lambda and 4.1 pp for Mimivirus. This GC gap later proved to be the most analytically productive architectural feature, as it reflects the AT-enriched regulatory signals concentrated in intergenic space.

Table 1. Genome architecture across the four-genome comparison ladder.

Metric	PhiX174	Lambda	Mimivirus	<i>P. salinus</i>
Genome size (bp)	5,386	48,502	1,181,549	2,473,870
GC content	44.7%	49.9%	28.0%	61.7%
Coding fraction	100.0%	88.2%	90.4%	67.5%

Metric	PhiX174	Lambda	Mimivirus	<i>P. salinus</i>
Coding GC	44.7%	51.1%	28.7%	64.6%
Intergenic GC	N/A	44.1%	24.6%	53.9%
GC gap (coding - IG)	N/A	7.0 pp	4.1 pp	10.7 pp
Avg gene length (bp)	490	665	1,058	1,168
Strand ratio (+/total)	1.000	0.644	0.488	0.504

Null model calibration using Markov-1 models preserving dinucleotide frequencies revealed two sequence features of particular note. First, poly-A/T hexamers (AAAAAA/TTTTTT) are 15-fold enriched relative to null expectations ($z > +100$), confirming that homopolymer runs are genuine sequence features maintained by selection, not base composition artifacts. Second, the AGCT tetranucleotide is depleted 1,500-fold relative to Markov-1 expectations ($z = -82$), consistent with the extreme compositional constraint reported by Poirot et al. (2019). The effect-size filtering methodology proved essential: in a genome of 2.47 Mb, 92.6% of 4-mers pass $z > 3$ by statistical power alone, but adding an effect-size threshold (observed/expected < 0.5 or > 2.0) reduces this to 10.5%, demonstrating that 88.6% of nominally significant k-mer signals lack meaningful biological effect sizes.

Codon usage analysis revealed that *P. salinus* exhibits translational selection (neutrality plot $R^2 < 0.05$) with codons better adapted to the *Acanthamoeba* host translation machinery than Mimivirus (CAI 0.787 vs 0.657). Paradoxically, Mimivirus shows stronger codon bias (ENC 41.6 vs 45.6) despite worse host adaptation, possibly because its poorly matched tRNA complement forces concentration on a narrow subset of usable codons. Critically, ORFan genes were compositionally indistinguishable from annotated genes across all codon metrics: GC3 difference +0.015, ENC difference -1.03, CAI difference +0.008, and upstream 6-mer cosine similarity 0.9824 (Supplementary Table S1). If the 92.5% ORFan fraction had been recently acquired through horizontal gene transfer from diverse sources, these genes would carry the codon signatures of their origins. Instead, the compositional homogeneity indicates either ancient residency with full equilibration or de novo origin within the genome.

Ab initio gene prediction with Prodigal identified 1,685 CDS in *P. salinus*, confirming 93.4% of GenBank annotations ($\geq 80\%$ reciprocal overlap) while predicting 347 novel genes not in GenBank (mean length 380 bp, mean GC 57.0%). These novel predictions reduce the estimated “true” intergenic fraction from 32.5% to approximately 23.8%, suggesting the genome encodes more genes than currently annotated. The native zone (1.3-2.0 Mb), identified independently by convergence of a GC content dip in sliding-window profiles, a GC skew inflection, and corroborated by GMM clustering, showed a higher confirmation rate (96.0% vs 92.4%) and lower hypothetical fraction (55.2% vs 66.2%) than the rest of the genome which is better annotated, not a gene desert, and its compositional distinctiveness reflects the coding genes themselves.

GMM clustering on five per-gene features (GC, GC3, GC12, ENC, CAI) identified three compositional clusters: a main native cluster (53.8%, GC 0.651), a high-GC cluster (29.4%, GC 0.654), and a variant cluster (16.8%, GC 0.613). Only 10 genes (0.7%) had low-confidence cluster assignments. Fifty-one extreme low GC outliers (GC < 0.567) all hypothetical were concentrated in the variant cluster. The native zone showed significantly different cluster composition (chi-squared = 19.26, $p = 6.6 \times 10^{-5}$), with more native-cluster genes (63% vs 50%) and fewer high-GC genes (23% vs 32%). Despite this local variation, *P. salinus* exhibited the lowest GC coefficient of variation of all tested genomes (6.1% vs Lambda 14.2%, Mimivirus 15.1%; Supplementary Table S8), a remarkable result given that it is 460-fold larger than PhiX174. This genome-wide compositional uniformity is among the strongest evidence against recent patchwork assembly from diverse sources.

ESM-2 protein language model embeddings for 170 representative proteins spanning all compositional clusters and proto-gene tiers revealed 9 structural families, in contrast to the 14 compositional families detected by protein 3-mer similarity (Module 7, covering only 3.2% of genes). Null calibration against composition-matched random proteins confirmed that real structural families are significantly tighter than expected by chance (within-family similarity 0.533 vs null 0.368; Cohen's $d = 1.45$), although 41% of PC1 variance correlates with GC content, indicating a partial compositional confound. Module 7 compositional families were validated structurally: 10 of 14 family representatives clustered together alongside known domains (Fascin, ankyrin, kinase, Ubiquitin). Twenty-three ORFan singletons clustered structurally with annotated domain-containing proteins, suggesting deeper homology relationships invisible to sequence-based methods (Supplementary Table S2). Eight of nine structural families were cross-tier, indicating that structural relationships cut across compositional categories proteins with different GC content, annotation status, and cluster assignments share structural embedding features.

A five-tier proto-gene continuum

Gene family analysis of the *P. salinus* proteome revealed that 96.8% of genes are compositional singletons with no detectable similar partners at the 0.50 cosine similarity threshold. The 14 detected families involve domain-repeat expansions (Fascin-like: 12 members; ankyrin: 8, 4, and 2 members) rather than whole-gene duplications, and show no tandem duplication signal family members are dispersed across the genome. Null calibration confirmed that the elevated background pairwise similarity (mean 0.109 vs Mimivirus 0.069) is fully explained by *P. salinus*'s skewed amino acid composition driven by 62% GC (null mean 0.105; Cohen's $d = 0.06$), while the families themselves rise above a well-characterized null background (Supplementary Table S3). These results disfavor large-scale gene duplication as the primary genome growth mechanism.

Classification of all 2,292 predicted ORFs (1,430 GenBank + 862 additional Prodigal predictions) into five confidence tiers revealed a monotonic gradient across every measured compositional feature (Table 2). GC content declined continuously from 0.650 in mature annotated genes (T1) to 0.547 in the most immature proto-gene candidates (T5) approaching the intergenic average of 0.539. Gene length followed the same trajectory

(1,452 bp -> 216 bp), as did GC3 (0.750 -> 0.672 for the tiers where wobble position could be computed) and CAI (0.792 -> 0.762).

Table 2. Five-tier proto-gene continuum. Spearman rho = -1.0 for both GC and length across tiers.

Tier	Description	n	Mean length (bp)	Mean GC	GC3	CAI
T1	Annotated, Prodigal-confirmed	516	1,452	0.650	0.750	0.792
T2	Hypothetical, Prodigal-confirmed	784	1,082	0.648	0.745	0.787
T3	GenBank-only (Prodigal missed)	130	555	0.615	0.672	0.762
T4	Prodigal high-confidence only	335	501	0.560	–	–
T5	Prodigal low-confidence only	527	216	0.547	–	–

The gradient did not merely reflect annotation quality. Codon usage analysis comparing T4 and T5 predictions against length-matched random intergenic ORFs revealed statistically significant genome-directed optimization even in the least mature candidates: T4 codon usage was significantly closer to the genome-preferred pattern than random ($p = 0.020$), with T5 showing an even stronger signal ($p = 0.0005$). The effect size was modest approximately 10-15% of the distance from random to fully optimized consistent with weak but detectable selection on recently emerged ORFs. Prodigal does not incorporate codon preference in its prediction algorithm, so this signal must originate from the sequences themselves.

Proto-genes showed no spatial clustering near established genes (T4 $p = 0.99$; T5 $p = 1.0$), ruling out regulatory co-option or operon extension as mechanisms. Instead, proto-gene candidates were distributed uniformly across the genome, consistent with a model in which gene birth can occur anywhere in the intergenic landscape.

Proto-genes were also not structurally isolated from established genes. ESM-2 nearest-neighbor similarity for proto-gene candidates (0.795) was comparable to established genes (0.751; Mann-Whitney $p = 0.90$), and the proto-gene centroid fell within the established gene distribution in PCA space (centroid distance/spread ratio = 0.81). Twelve of 15 proto-gene candidates clustered structurally with the low-GC outlier group, consistent with early evolutionary stage structurally simple, low-GC, still acquiring the features of mature proteins. This structural non-separability suggests continuity in protein-level organization rather than arbitrary ORF detection, though it is partially confounded by

205 shared amino acid composition between low-GC proteins (see Genome-wide
206 compositional landscape above).

207 A distributed AT-based regulatory system

208 The most striking finding of the compositional analysis was the discovery of an AT-based
209 regulatory system operating against the genome's high GC content. After filtering to
210 intergenic runs only (see Transcriptomic validation below), *P. salinus* exhibited gene
211 boundary AT signals dramatically stronger than any other genome tested: +10.3
212 percentage points AT enrichment at gene starts and +9.8 percentage points at gene ends,
213 compared to approximately +1 percentage point for Lambda and +1-2 percentage points
214 for Mimivirus. This signal runs counter to the genome's 62% GC composition, requiring
215 active regulatory selection to maintain. The full cross-genome comparison of AT boundary
216 signals is provided in Supplementary Table S10.

217 Poly-A and poly-T homopolymer runs (≥ 5 bp) numbered 6,955 and 7,010 respectively,
218 with 82.8-83.9% located in intergenic regions a pattern unique to *P. salinus* (Lambda: 12-
219 24%; Mimivirus: 16-17%). These runs displayed strand-asymmetric positioning consistent
220 with regulatory function: poly-A runs were enriched upstream of gene starts (41.5% within
221 200 bp), while poly-T runs were enriched downstream of gene ends (40.8%).

222 Divergent intergenic regions (flanked by genes transcribed outward in both directions)
223 were 17% longer than convergent regions (582 bp vs 498 bp; $p < 0.0001$), consistent with a
224 requirement for two promoters to be accommodated. However, divergent regions were
225 less dense in poly-A runs (6.67 vs 7.69 runs/kb; $p = 0.019$), indicating that the extra length
226 represents spacer DNA between two normal-density promoter regions.

227 The critical finding for the de novo gene birth model was that proto-gene candidates
228 exhibited regulatory signals indistinguishable from established genes. T4 upstream AT
229 content was 45.0% ($p = 0.14$ vs annotated 45.9%) and T5 was 46.7% ($p = 0.25$) neither
230 significantly different from mature genes. This means that the regulatory infrastructure for
231 transcription pre-exists in intergenic space before ORFs emerge, addressing the
232 mechanistic gap of how spatially random proto-genes can be expressed: they are born into
233 a landscape already equipped with promoters and terminators.

234 Transcriptomic validation

235 To validate the computationally predicted regulatory signals, we compared poly-A/T run
236 positions against experimentally informed gene annotations from Legendre et al. (2018).
237 After filtering to 11,640 intergenic runs and applying strand-aware classification, we found
238 that 74% of boundary-associated runs occupied the correct regulatory position poly-A
239 upstream on the sense strand or poly-T upstream on the antisense strand for promoters,
240 and the converse for terminators (binomial $p < 10^{-6}$).

241 This strand-aware analysis identified 2,332 runs in promoter-consistent positions and
242 1,227 in terminator-consistent positions. The asymmetry (1.9:1 promoter-to-terminator

ratio) may reflect the greater complexity of transcription initiation relative to termination, or may indicate that a subset of termination events uses the hairpin mechanism described by Legendre et al. (2018) rather than poly-T signals.

Meta-gene AT content profiles averaging AT content across aligned gene boundaries revealed sharp spikes at transcript boundaries: +10.3 percentage points at gene starts and +9.8 percentage points at gene ends. These values exceed the initial unfiltered estimates (+7.0/+9.5 pp from Module 9) because the intergenic filter removed coding-region runs that diluted the signal.

To assess whether palindromic hairpin structures, proposed by Legendre et al. (2018) as the primary termination signal, could account for the poly-A/T signal, we implemented a hairpin null model with strict parameters (stem ≥ 6 bp, exact complementarity). The observed hairpin fraction (11.6%) exceeded the null model expectation (9.7%) by only 2.0 percentage points ($z = 3.6$, statistically significant but biologically small). The 70% hairpin frequency reported by Legendre et al. (2018) likely reflects energy-based RNA folding analysis (e.g., RNAfold/mfold) rather than simple palindrome detection. Importantly, the poly-A/T signal stands independently of hairpin structures.

The two-population test comparing boundary-proximal ($n = 4,331$) and dispersed intergenic ($n = 7,309$) poly-A/T runs found no compositional difference in AT ratio ($p = 0.105$) or length ($p = 0.093$). This supports the latent regulatory substrate model: dispersed runs throughout intergenic space are compositionally identical to those actively functioning at gene boundaries. The regulatory potential is distributed, not concentrated any region of the genome could, in principle, support transcription of a newly emerged ORF.

Homology landscape and the maturation gradient

BLASTp analysis of all 528 annotated *P. salinus* proteins against the NCBI nr database revealed that 426 of 521 successful queries (81.8%) returned hits exclusively to other *Pandoravirus* strains, confirming the extreme lineage-specificity of this proteome. The 95 proteins with non-self hits showed an eukaryote-dominated taxonomic distribution (Table 3): marine algae (28, 29.5%), giant viruses (24, 25.3%), Amoebozoa (13, 13.7%), bacteria (11, 11.6%), insects (11, 11.6%), other eukaryotes (7, 7.4%), and other viruses (1, 1.1%).

Table 3. Taxonomy and identity of non-self BLAST hits (annotated genes, $n = 95$).

Category	Count	%	Mean identity
Marine algae (prasinophytes + brown algae)	28	29.5	44.2%
Giant virus (NCLDV)	24	25.3	47.5%
Amoebozoa (host lineage)	13	13.7	50.1%
Bacteria	11	11.6	38.1%
Insect (aphids, likely endosymbiont-mediated)	11	11.6	41.1%
Other eukaryote	7	7.4	42.9%

Category	Count	%	Mean identity
Other virus	1	1.1	–

The marine algae category represents an ecological grouping, not a phylogenetic one: green algae (Chlorophyta/Prasinophyceae; *Micromonas*, *Bathycoccus*, *Ostreococcus*) and brown algae (Phaeophyceae; *Ectocarpus*, *Scytosiphon*) are as distantly related as animals and fungi, yet both share marine coastal habitat with *P. salinus*. The 11 insect hits mapped exclusively to ankyrin and F-box domains in aphid genomes (*Acyrtosiphon pisum*, *Aphis* spp.); these likely reflect convergent domain expansion mediated by Wolbachia endosymbionts rather than direct Pandoravirus-to-insect horizontal transfer.

Non-self hits were spatially enriched in the native zone (1.3-2.0 Mb): 45 of 95 hits fell within this region (OR = 1.94, Fisher's exact $p = 0.004$), consistent with the native zone containing evolutionarily older genes that retain broader phylogenetic connections. Product-level analysis revealed a gradient from clearly acquired domains (ankyrin repeat: 56/132 = 42.4% non-self hit rate) through diverged domains (F-box: 10/118 = 8.5%, MORN repeat: 4.5%) to Pandoravirus-specific domains with zero external homologs (Fascin, Atrophin, DHFR, Ring finger, BTB: 0%), providing direct evidence for different evolutionary ages of domain acquisition (Supplementary Table S4).

To test whether homology tracks gene maturity the central prediction of the maturation model we compared annotated and hypothetical proteins directly. Of 299 successfully queried hypothetical proteins, only 9 (3.0%) had non-self hits, compared to 95 of 521 (18.2%) for annotated proteins a 7.2-fold enrichment (Fisher's exact test, OR = 7.19, 95% CI: 3.5-13.6, $p = 7.79 \times 10^{-12}$; Table 4). This result was robust across E-value thresholds: the annotated orphan rate remained above 80% even at the most stringent threshold tested (81.8% at 10^{-5} , 83.1% at 10^{-10} , 84.1% at 10^{-20} , 90.4% at 10^{-50} ; Supplementary Table S5).

Table 4. Annotated vs hypothetical homology comparison.

Gene category	n	Non-self hit rate	Interpretation
Annotated (recognized domains)	52 1	18.2%	Oldest genes; some retain detectable homology
Hypothetical (ORFs, no domains)	29 9	3.0%	Younger genes; almost entirely orphans
Intergenic regions	~1, 90 0 IG Rs	0% (by definition)	Not yet genes; the raw substrate

This comparison reveals not only a quantitative difference in orphan rate, but also a qualitative contraction in phylogenetic reach among younger genes. The 9 hypothetical non-self hits displayed a taxonomic profile distinct from annotated hits: 6 giant viruses (66.7%), 2 Amoebozoa (22.2%), and 1 bacterium (11.1%) with zero marine algae and zero

insects. Hypothetical genes with detectable external homology connect exclusively to phylogenetically proximate sources (the NCLDV gene pool and the amoebozoan host lineage), while more distant connections (marine algae, insects, diverse bacteria) appear only in annotated genes. This phylogenetic distance gradient younger genes retain similarity only to nearest relatives is an additional prediction of the maturation model confirmed by the data.

Discussion

A mechanistic model for de novo gene birth

The results presented here converge on a coherent model for de novo gene birth in *P. salinus* that can be articulated as a three-step logical chain. First, poly-A/T runs with regulatory positioning exist throughout the intergenic space, as demonstrated by the 82.8-83.9% intergenic fraction, the strand-asymmetric boundary enrichment, and the +10.3/+9.8 pp meta-gene AT spikes confirmed by strand-aware validation. Second, boundary-proximal and dispersed intergenic runs are compositionally identical (AT ratio $p = 0.105$, length $p = 0.093$), indicating that the regulatory substrate is distributed uniformly rather than concentrated at existing gene boundaries. Third, proto-genes emerge into these pre-existing regulatory contexts with mature transcriptional signals already in place (T4 $p = 0.14$, T5 $p = 0.25$), acquire coding features through a gradual maturation process measurable as the five-tier continuum, and progressively accumulate external homologs as they age (3.0% $\rightarrow$ 18.2% non-self hit rate).

This model predicts a specific maturation ordering. Regulatory signals pre-exist as a feature of intergenic sequence composition, representing the earliest (indeed, prerequisite) stage. Protein structural features emerge next proto-genes were not structurally separable from established domain families in embedding space (nearest-neighbor similarity test, $p = 0.90$), consistent with early acquisition of protein-level organization even among the most immature candidates. Codon optimization follows, detectable as a 10-15% shift from random toward genome-preferred usage (T4 $p = 0.020$, T5 $p = 0.0005$). GC composition shifts last, as the slowest and most passive process, with proto-genes (GC 0.547) only gradually approaching the coding average (GC 0.650) from the intergenic baseline (GC 0.539).

Evidence against alternative genome growth models

Multiple independent lines of evidence converge against the two standard alternative models for genome expansion. Against recent patchwork horizontal gene transfer: *P. salinus* exhibits the lowest GC coefficient of variation (6.1%) of any genome tested despite being the largest; ORFan genes are compositionally indistinguishable from annotated genes across all codon metrics (GC3 difference +0.015, CAI difference +0.008); upstream regulatory signatures are shared (6-mer cosine similarity 0.9824); GMM clustering finds no dramatically foreign compositional outliers; and the BLASTp taxonomy shows an

eukaryote-dominated pattern consistent with shared ecology rather than multi-kingdom HGT. Against recent gene duplication: 96.8% of genes are compositional singletons; the 14 detected families involve domain-repeat expansions, not whole-gene duplications; family members are dispersed across the genome with no tandem clustering; ORFan genes show no tendency to form compositional families; and the diffuse pairwise similarity cloud (mean 0.109) is fully explained by a null model (Cohen's $d = 0.06$). These results leave de novo gene birth from non-coding sequence as the most parsimonious explanation, consistent with the comparative genomic evidence from the Claverie group.

An alternative interpretation is that the T4 and T5 proto-gene tiers reflect annotation artifacts or false-positive Prodigal predictions rather than genuine evolutionary intermediates. Several observations argue against this. First, both T4 and T5 exhibit codon usage significantly shifted toward genome-preferred patterns relative to length-matched random intergenic ORFs, with the strongest signal retained even in T5 ($p = 0.0005$), indicating sequence-level structure inconsistent with random noise. Second, representative proto-genes are not structurally isolated but cluster near established domain families in embedding space, suggesting continuity in protein-level organization rather than arbitrary ORF detection. Third, proto-genes show no spatial clustering near established genes, arguing against simple operon extension or boundary spillover from neighboring annotations. Taken together, these results are more consistent with a continuum of gene maturation than with a sharp division between “real genes” and prediction noise.

Relationship to existing work

Our findings provide an independent computational line of support for the de novo gene birth model proposed by Legendre et al. (2018) and Jeudy et al. (2019), arrived at through a fundamentally different methodology single-genome compositional analysis rather than multi-strain comparative genomics. The convergence of these independent approaches substantially strengthens the de novo gene birth hypothesis.

Our poly-A/T compositional analysis and the RNA secondary structure analysis of Legendre et al. (2018) may detect different aspects of the same termination system, operating at the sequence composition level and the RNA folding level respectively. The hairpin null model results (2.0 pp excess over null) suggest that simple palindrome detection captures only a fraction of the termination signal that energy-based RNA folding methods detect, and that the poly-A/T and hairpin mechanisms are complementary rather than competing.

The homology landscape reveals three distinct ecological narratives within the non-self hit taxonomy. The NCLDV connection (24 hits, 25.3%) represents the ancestral viral gene pool shared within the Nucleocytoviricota. The marine ecological network (28 algae + 13 Amoebozoa = 41 hits, 43.2%) reflects gene exchange within a shared coastal aquatic habitat notably involving two phylogenetically distant algal lineages (green and brown algae), strengthening the ecological rather than phylogenetic interpretation. The

endosymbiont signal (11 insect + 11 bacteria = 22 hits, 23.2%) is dominated by ankyrin repeats and likely reflects a shared bacterial HGT reservoir: aphid genomes contain many ankyrins acquired from *Wolbachia* endosymbionts, making the Pandoravirus-aphid connection indirect rather than direct.

The product-level gradient from ankyrin (42.4% non-self hit rate) through F-box (8.5%) to Fascin and Atrophin (0%) provides a temporal dimension: domains with high hit rates were likely acquired from external sources and retain recognizable homologs, while domains with zero external hits have either diverged beyond recognition or originated de novo within the *Pandoravirus* lineage.

Additional compositional forces

The intergenic space from which new genes emerge is shaped by multiple forces beyond the AT regulatory system described here. Poirot et al. (2019) documented the near-complete absence of the AGCT tetranucleotide in *Pandoravirus* genomes, which we independently confirmed as a 1,500-fold depletion relative to Markov-1 null expectations ($z = -82$). Survey of all 16 palindromic 4-mers revealed a mixed pattern 8 depleted, 7 enriched, 1 neutral with AT-containing palindromes depleted and GC-rich palindromes enriched, consistent with a compositional pattern driven by the high-GC genome rather than a specific restriction-modification system. Whether this reflects a novel DNA editing process or extreme compositional pressure, it demonstrates that *Pandoravirus* intergenic sequence is subject to selective constraints beyond neutral evolution. Sun et al. (2015) showed that MITE transposable elements colonize *P. salinus* intergenic space, raising questions about possible interactions between mobile elements and the distributed regulatory signals we describe. Whether MITEs disrupt, co-opt, or are recruited alongside AT-rich regulatory elements remains to be determined.

Limitations

This study has several important limitations. All findings derive from a single genome (*P. salinus*); generalizability to other *Pandoravirus* strains and to other Nucleocyotviricota awaits confirmation. The transcriptomic validation used GFF annotations informed by RNA-seq data rather than raw read-level TSS/TTS coordinates, which are not publicly available; direct comparison of predicted regulatory signals against experimentally determined transcript boundaries would provide stronger validation. ESM-2 embeddings are a structural proxy, not actual 3D structure predictions; validation with AlphaFold or experimental structure determination would strengthen the structural maturation findings, and 41% of the primary structural axis correlates with GC content. The hypothetical gene subsample ($n = 299$) represents approximately one-third of all hypothetical genes; complete coverage would further refine the orphan rate estimates. Compositional methods reveal evolutionary history but cannot determine gene expression, functionality, or essentiality the 92.5% ORFan fraction remains functionally uncharacterized. The CAI reference used general *A. castellanii* codon frequencies rather than a highly-expressed gene subset; CAI values are valid for relative comparisons but not absolute measures of

translational optimization. Finally, the analysis pipeline was developed with AI tool assistance (see Use of AI Tools); while all results were independently verified, the analytical design reflects a human-AI collaborative process that may have introduced unrecognized biases in the choice of methods or interpretation of results.

Predictions

Our model generates four falsifiable predictions. First, comparative genomic analysis of *P. dulcis* and *P. neocaledonia* should reveal proto-genes at different maturation stages, with some genes present in one strain corresponding to intergenic regions in others (partially confirmed by Jeudy et al. 2019 for *P. celtis*/*P. quercus*). Second, the strand-aware positional signal is consistent with promoter/terminator function; direct validation against experimentally determined TSS/TTS sites awaits reanalysis of primary transcriptomic data. Third, the 23 ORFan singletons that cluster structurally with known domain families (Fascin, ankyrin, kinase, Ubiquitin) should show structural similarity when analyzed with AlphaFold or experimentally determined. Fourth, we predict that poly-A/T runs in intergenic regions adjacent to hypothetical genes will show promoter or terminator activity in a reporter assay, even when the adjacent ORF is not conserved across *Pandoravirus* strains providing a direct experimental test of the distributed regulatory infrastructure model.

Materials and Methods

Genome and database

The complete genome of *Pandoravirus salinus* (GenBank accession NC_022098.1; 2,473,870 bp, 1,430 annotated CDS) was loaded into a PostgreSQL 15 database alongside three comparison genomes spanning the dsDNA virus size spectrum: bacteriophage PhiX174 (NC_001422.1; 5,386 bp), bacteriophage Lambda (NC_001416.1; 48,502 bp), and *Acanthamoeba polyphaga* mimivirus (NC_014649.1; 1,181,549 bp). The database schema comprises 22 tables including position-indexed nucleotide sequences, gene annotations, intergenic regions, codon usage statistics, and derived analytical tables. All analyses used Python 3.11 with BioPython 1.86, NumPy, SciPy, scikit-learn, and matplotlib.

Compositional and structural analysis pipeline (Modules 2-8)

A 12-module analytical pipeline was applied to characterize the compositional and structural landscape of *P. salinus*. Here we summarize the methods underlying findings presented in Results; full pipeline details and scripts are available in the Supplementary Materials.

Genome architecture (Module 2). Eight metrics were computed genome-wide and per-gene: GC content, coding density, strand bias, intergenic region length, Shannon entropy, gene length distribution, and sliding-window GC profiles. All metrics were compared across the four-genome ladder.

Null model calibration (Module 3). Markov-1 null models preserving dinucleotide frequencies were used to assess k-mer enrichment and depletion ($k = 4, 6, 8$). Effect-size filtering (observed/expected < 0.5 or > 2.0) was applied alongside z-score significance to reduce false positives in large genomes, where 88.6% of $z > 3$ signals lacked meaningful effect sizes.

Codon usage analysis (Module 4). Per-gene relative synonymous codon usage (RSCU), GC3 (wobble position), GC12 (constrained positions), effective number of codons (ENC), and codon adaptation index (CAI) were computed. CAI was calculated using *Acanthamoeba castellanii* codon frequencies from the Kazusa database. Annotated versus ORFan gene comparisons tested whether ORFans show distinct compositional signatures indicative of foreign acquisition.

Gene prediction validation (Module 5). Ab initio gene prediction with Prodigal was compared against GenBank annotations using $\geq 80\%$ reciprocal overlap matching. Novel Prodigal-only predictions ($n = 347$) were characterized by length, GC content, and genomic position.

Evolutionary origin classification (Module 6). Gaussian Mixture Model (GMM) clustering on five per-gene features (GC, GC3, GC12, ENC, CAI) identified compositional clusters. The native zone (1.3-2.0 Mb), defined independently by this clustering, was used in subsequent spatial analyses.

Gene family and proto-gene analysis (Modules 7-7c). All-vs-all protein 3-mer frequency cosine similarity (1,021,735 pairwise comparisons) with hierarchical clustering identified gene families. Null calibration using composition-matched random proteins confirmed that the diffuse similarity cloud (mean 0.109) is a compositional artifact (null mean 0.105; Cohen's $d = 0.06$), while the 14 detected families represent genuine signals.

Proto-gene candidates were classified into five confidence tiers (T1-T5) based on prediction agreement between GenBank annotations and Prodigal: T1, annotated genes confirmed by both ($n = 516$); T2, hypothetical genes confirmed by both ($n = 784$); T3, GenBank-only genes missed by Prodigal ($n = 130$); T4, high-confidence Prodigal-only predictions ($n = 335$); T5, low-confidence Prodigal-only predictions ($n = 527$). Disambiguation tests compared T4/T5 codon usage against length-matched random intergenic ORFs and assessed spatial proximity to established genes.

Protein structure prediction (Module 8). ESM-2 (650M parameters; Lin et al. 2023) embeddings were generated for 170 representative proteins spanning all tiers and compositional clusters. PCA reduction (top 20 components, 85.8% variance) followed by hierarchical clustering identified structural families. Null calibration confirmed structural signal genuineness (within-family similarity 0.533 vs null 0.368; Cohen's $d = 1.45$), though 41% of PC1 variance correlates with GC content.

493 Regulatory signal analysis (Module 9)

494 Seven complementary analyses characterized the regulatory landscape: (i) genome-wide
495 poly-A/T run inventory (runs ≥ 5 bp), (ii) meta-gene AT content profiles across gene
496 boundaries, (iii) upstream k-mer enrichment analysis, (iv) intergenic grammar comparison
497 between convergent and divergent regions, (v) strand-asymmetric boundary positioning, (vi)
498 ORFan versus annotated regulatory signature comparison, and (vii) proto-gene regulatory
499 context assessment.

500 Transcriptomic validation (Module 10)

501 Computationally predicted regulatory signals were compared against experimentally
502 informed gene annotations from Legendre et al. (2018). GFF annotations incorporating
503 RNA-seq evidence were used in lieu of raw reads, which are not deposited in public
504 repositories.

505 Three methodological corrections were applied relative to an initial analysis: (i) an
506 intergenic filter removed 2,325 poly-A/T runs (16.6%) overlapping coding sequences,
507 retaining 11,640 intergenic runs; (ii) a strand-aware boundary test classified each run as
508 promoter-positioned (poly-A upstream of sense-strand gene start, or poly-T upstream of
509 antisense-strand gene start) or terminator-positioned, replacing a strand-naive proximity
510 test; (iii) a hairpin null model using strict parameters (stem ≥ 6 bp, exact complementarity)
511 quantified the contribution of palindromic sequences to termination signals.

512 The two-population test compared AT composition and length distributions between
513 boundary-proximal runs (within 200 bp of a gene boundary; $n = 4,331$) and dispersed
514 intergenic runs ($n = 7,309$) to test whether boundary-active and dispersed runs represent
515 distinct populations or a uniform regulatory substrate.

516 BLASTp homology analysis (Modules 11-12)

517 All 528 annotated (non-hypothetical) *P. salinus* protein sequences were searched against
518 the NCBI non-redundant protein database (nr) using remote BLASTp via BioPython (E-value
519 threshold $1e-5$, top 10 hits per query). Of 528 queries, 521 returned valid results; 7
520 connection failures were excluded from analysis. A subsample of 300 randomly selected
521 hypothetical proteins (from 902 total; random seed 42 for initial 100, seed 43 for additional
522 200) was searched under identical parameters, yielding 299 successful queries.

523 Non-self hits were defined as hits to organisms other than *Pandoravirus* strains.
524 Taxonomic classification used a curated organism-level lookup table covering all 50
525 unique organisms encountered in the dataset, supplemented by an ordered keyword
526 fallback and NCBI Entrez API lookup for unrecognized organisms. During quality auditing,
527 the initial automated classification was found to have misidentified 12 prasinophyte green
528 algae as bacteria due to a substring match on the suffix '-monas'; this was corrected by
529 implementing the curated lookup table approach.

Identity distribution, spatial enrichment in the native zone (1.3-2.0 Mb), and product-level hit rates were computed. The annotated versus hypothetical comparison used Fisher's exact test with Haldane-corrected Woolf method for the 95% confidence interval on the odds ratio, and Wilson score intervals for proportions. E-value sensitivity analysis re-filtered existing hits at thresholds of 10^{-10} , 10^{-20} , and 10^{-50} to assess robustness.

Use of AI Tools

Artificial intelligence tools (Anthropic's Claude) were used to assist with code development, data-processing workflow design, analytical planning, and editorial refinement of manuscript text. All analyses, statistical methods, interpretations, and final manuscript content were reviewed and validated by the author.

Author Contributions

E.B. conceived the study, designed and implemented the analytical pipeline, performed all computational analyses, and wrote the manuscript.

Data Availability

All analysis scripts, the PostgreSQL database schema, intermediate results, and raw BLAST output are available at <https://github.com/mathamagician/pandoravirus-denovo-genebirth> and archived with a permanent DOI at <https://doi.org/10.5281/zenodo.19046142>. The *P. salinus* genome sequence is available under GenBank accession NC_022098.1. RNA-seq-informed GFF annotations were obtained from Legendre et al. (2018) via the IGS GBrowse server (<http://www.igs.cnrs-mrs.fr/pandoraviruses/>). A statistics audit file mapping every manuscript-cited number to its source computation is included in the repository.

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573

574 Figure Legends

575 **Figure 1. Five-tier proto-gene continuum and maturation ordering.** (A) GC content
 576 across five confidence tiers, showing monotonic decline from T1 (0.650) to T5 (0.547), with
 577 the intergenic average (0.539) indicated by a dashed line. (B) Gene length distribution
 578 across tiers (box plots showing median, interquartile range, and outliers). (C) Codon usage
 579 displacement of T4 and T5 proto-genes relative to random intergenic ORFs and genome-
 580 optimized annotated genes (T1+T2), showing 10-15% displacement toward genome-
 581 preferred codons. (D) Schematic of the four-stage maturation model: regulatory signals
 582 (pre-existing) -> protein structure -> codon optimization -> GC composition.

583 Alt text: Four-panel figure showing the proto-gene continuum. Panel A is a bar chart of
 584 mean GC content declining from tier 1 (0.650) to tier 5 (0.547). Panel B shows box plots of
 585 gene length decreasing across tiers. Panel C is a bar chart showing codon usage
 586 displacement of proto-genes between random and genome-optimized levels. Panel D is a
 587 schematic diagram with four sequential arrows showing the maturation order from
 588 regulatory signals through protein structure, codon optimization, and GC composition.

589 **Figure 2. Distributed AT regulatory system and transcriptomic validation.** (A) Meta-
 590 gene AT content profile showing +10.3 pp spike at gene starts and +9.8 pp spike at gene
 591 ends. (B) Strand-aware classification of poly-A/T runs: 74% in correct regulatory position
 592 (binomial $p < 10^{-6}$). (C) Comparison of AT boundary signal strength across Lambda,
 593 Mimivirus, and *P. salinus*. (D) Two-population test comparing boundary-proximal runs
 594 (within 200 bp of a gene boundary, $n = 4,331$) against dispersed intergenic runs ($n = 7,309$):
 595 no significant difference in AT composition between the two groups (AT ratio $p = 0.105$),
 596 supporting a uniform regulatory substrate. (E) Proto-gene regulatory signals: T4 and T5
 597 upstream AT content indistinguishable from annotated genes. (F) Hairpin null model:
 598 observed (11.6%) vs null (9.7%), 2.0 pp excess.

599 Alt text: Six-panel figure showing the AT regulatory system. Panel A is a line plot of AT
 600 content across gene boundaries showing sharp upward spikes at start and end positions.
 601 Panel B is a stacked bar chart showing 74% of poly-A/T runs in correct strand-aware

regulatory position. Panel C is a grouped bar chart comparing AT boundary signal strength across three genomes, with *P. salinus* showing dramatically larger signals. Panel D shows overlapping density plots for boundary-proximal and dispersed poly-A/T runs with nearly identical distributions. Panel E is a bar chart comparing upstream AT content across gene tiers showing no significant differences. Panel F is a bar chart comparing observed and null-model hairpin frequencies.

Figure 3. BLASTp homology landscape. (A) Taxonomic distribution of non-self top BLAST hits ($n = 95$), showing eukaryote-dominated pattern with marine algae (29.5%) and giant viruses (25.3%) as the largest categories. (B) Percent identity distribution of non-self hits (median 42.3%). (C) Product-level non-self hit rates, showing gradient from ankyrin (56/132 = 42.4%) through F-box (10/118 = 8.5%) to Pandoravirus-specific domains (Fascin, DHFR, Atrophin, Ring finger: 0%). Counts shown as hits/total for each product category. (D) Spatial distribution of non-self homology hits across the genome, with points colored by taxonomy and the native zone (1.3-2.0 Mb, shaded) highlighted; this region shows enrichment for non-self hits (45/95, OR = 1.94, $p = 0.004$). (E) Identity distribution by taxonomic category.

Alt text: Five-panel figure showing the BLASTp homology landscape. Panel A is a pie chart of taxonomic categories of non-self hits dominated by marine algae and giant viruses. Panel B is a histogram of percent identity values centered around 42%. Panel C is a horizontal bar chart showing product-level hit rates declining from ankyrin at 42.4% to zero for several Pandoravirus-specific domains. Panel D is a scatter plot of hit positions across the genome with a shaded native zone region showing enrichment. Panel E shows box plots of identity distributions grouped by taxonomic category.

Figure 4. Maturation gradient: annotated vs hypothetical homology. (A) Non-self hit rates for annotated (18.2%, $n = 521$) versus hypothetical (3.0%, $n = 299$) genes, with Wilson score confidence intervals. Fisher's exact test OR = 7.19 (95% CI: 3.5-13.6, $p = 7.79 \times 10^{-12}$). The single highest-identity hypothetical hit (*Acanthamoeba castellanii*, 82.4% identity) is indicated. (B) Three-level maturation gradient: intergenic (0%) -> hypothetical (3.0%) -> annotated (18.2%). (C) E-value sensitivity analysis showing orphan rate stability across thresholds (81.8% at 10^{-5} to 90.4% at 10^{-50}).

Alt text: Three-panel figure showing the maturation gradient. Panel A is a bar chart comparing non-self hit rates between annotated (18.2%) and hypothetical (3.0%) genes with error bars showing confidence intervals. Panel B is a step chart showing the three-level gradient from intergenic (0%) through hypothetical (3.0%) to annotated (18.2%). Panel C is a line plot showing orphan rates remaining stable (above 80%) across increasingly stringent E-value thresholds.

Supplementary Materials

Supplementary Tables S1-S10 and Supplementary Figures S1-S6 are provided in a single supplementary PDF file.

Supplementary Table S1. Codon usage comparison between annotated and ORFan genes: RSCU, GC3, GC12, ENC, and CAI with statistical tests.

Supplementary Table S2. ESM-2 structural family assignments for 170 representative proteins, including tier, compositional cluster, and nearest-neighbor similarity.

Supplementary Table S3. Gene family detection results: all 14 compositional families with member counts, mean similarity, and spatial distribution.

Supplementary Table S4. Product-level non-self BLASTp hit rates for all annotated protein categories.

Supplementary Table S5. E-value sensitivity analysis for annotated gene non-self hit rates across four thresholds.

Supplementary Table S6. Complete taxonomy of all 95 non-self BLAST hits with organism, identity, coverage, and genomic position.

Supplementary Table S7. All 9 hypothetical gene non-self hits with taxonomy, identity, and native zone membership.

Supplementary Table S8. Cross-genome compositional homogeneity: full GC, GC3, GC12, ENC, and CAI coefficient of variation comparison.

Supplementary Table S9. Hypothesis testing matrix: evidentiary status of four genome growth models (HGT, gene duplication, mobile element expansion, de novo gene birth) across all 12 modules, with supporting/disfavoring classification and key statistics for each.

Supplementary Table S10. Gene boundary AT signals across genomes: AT content, start and end delta-AT values, and poly-A/T intergenic fraction for Lambda, Mimivirus, and *P. salinus*.

Supplementary Figure S1. Genome architecture: sliding-window GC profiles, gene density maps, GC skew plots, and strand maps for all four genomes in the comparison ladder.

Supplementary Figure S2. GMM clustering of *P. salinus* genes: three compositional clusters with native zone (1.3-2.0 Mb) overlay, showing cluster composition shift within the zone.

Supplementary Figure S3. Pairwise protein 3-mer cosine similarity distributions: *P. salinus* vs Mimivirus, with composition-matched null model overlay demonstrating that the

673 diffuse similarity cloud (mean 0.109) is a compositional artifact (null mean 0.105; Cohen's
674 $d = 0.06$).

675 **Supplementary Figure S4.** ESM-2 PCA embedding space showing structural family
676 assignments, proto-gene positions, GC content overlay, and null calibration comparison
677 (within-family similarity 0.533 vs null 0.368; Cohen's $d = 1.45$).

678 **Supplementary Figure S5.** Full 7-panel Module 11 BLASTp analysis figure (panels A-G).

679 **Supplementary Figure S6.** Module 10 corrected analysis 6-panel figure.